

WOMEN'S HEALTH, HRT & SEXUAL WELLNESS

Hormone Optimization · Sexual Health · Performance · Recovery

A science-backed reference guide covering female hormone replacement therapy, sexual health optimization, libido, arousal, menopause management, and evidence-based recommendations for women at every stage of life. Written for health-conscious women who want real answers based on data, not dismissal.

SECTION 1: FEMALE HORMONES — THE FOUNDATION

THE BIG FOUR: ESTROGEN, PROGESTERONE, TESTOSTERONE, DHEA

Hormone	Normal Range	Function	Deficiency Signs
Estradiol (E2)	Premenopause: 30–400 pg/mL Postmenopause: <30 pg/mL	Bone density, cardiovascular, vaginal health, libido, mood	Hot flashes, vaginal dryness, brain fog, osteoporosis
Progesterone	Luteal: 5–20 ng/mL Postmenopause: <1 ng/mL	Uterine health, sleep quality, anxiety regulation, libido	Disrupted sleep, anxiety, irregular cycles, estrogen dominance
Testosterone (Total)	15–70 ng/dL (women) Optimal: 30–70 ng/dL	Libido, muscle mass, energy, clitoral sensitivity	Low libido, fatigue, muscle loss, reduced orgasm intensity
DHEA-S	35–430 mcg/dL (age-dependent)	Precursor to estrogens and androgens, energy, mood	Low energy, mood decline, poor mood, accelerated aging

SECTION 2: HORMONE REPLACEMENT THERAPY (HRT)

THE EVIDENCE ON HRT — SETTING THE RECORD STRAIGHT

The 2002 WHI study caused decades of unnecessary HRT avoidance by overstating risks. Modern re-analysis shows that **bioidentical HRT started within 10 years of menopause (the 'timing hypothesis') significantly reduces cardiovascular disease, all-cause mortality, osteoporosis, and cognitive decline**. The risks of NOT treating menopause are now understood to be substantial.

BENEFITS OF TIMELY HRT

- Eliminates hot flashes and night sweats (90%+ relief)
- Restores vaginal moisture, elasticity, and pH — eliminates painful intercourse
- Improves sleep architecture and sleep quality significantly
- Reduces cardiovascular disease risk when started early (timing hypothesis)

- Preserves bone density — prevents osteoporosis fractures
- Reduces risk of type 2 diabetes and metabolic syndrome
- Improves mood, reduces depression and anxiety
- Cognitive protection — reduces Alzheimer's and dementia risk
- Restores libido and sexual function
- Improves skin collagen, moisture, and elasticity

HRT OPTIONS — ROUTES AND FORMULATIONS

Hormone	Route	Typical Dose	Notes
Estradiol	Transdermal patch	0.025–0.1 mg/day	Preferred route — avoids first-pass liver metabolism. Lower clot risk vs. oral.
Estradiol	Gel/cream (topical)	0.5–1.5 mg/day	Flexible dosing. Apply to inner arm or thigh daily.
Estradiol	Oral (Estrace)	0.5–2 mg/day	Convenient but first-pass metabolism — higher VTE risk vs. transdermal.
Progesterone	Oral (Prometrium)	100–200 mg at bedtime	Bioidentical. Improves sleep. Essential for uterine protection with estrogen.
Progesterone	Vaginal (Crinone)	4–8% gel	For uterine protection without systemic side effects.
Testosterone	Cream/gel (compounded)	0.5–2 mg/day	Applied to clitoris, labia, or inner thigh. Improves libido, orgasm.
Testosterone	Injection (IM/SC)	5–10 mg/week	More precise dosing for women with significant deficiency.
DHEA	Oral	25–50 mg/day	Precursor to both estrogen and testosterone. Monitor levels at 3 months.
DHEA (Intrarosa)	Vaginal suppository	6.5 mg/day	FDA approved for vaginal atrophy. Converts locally to estrogen + testosterone.

TESTOSTERONE THERAPY FOR WOMEN — THE MISSING PIECE

Female testosterone deficiency is dramatically underdiagnosed and undertreated. Women produce 3–4x more testosterone than estrogen — and its decline drives many of the symptoms women suffer silently: low libido, fatigue, muscle loss, depression, reduced orgasm intensity, and blunted motivation.

- Restores sexual desire and libido — most consistent evidence of any HRT modality
- Increases orgasm intensity and frequency
- Improves clitoral and vaginal sensitivity
- Increases lean muscle mass and reduces body fat
- Improves energy, motivation, and drive
- Reduces depression and anxiety independent of estrogen therapy
- Improves cognitive sharpness and working memory
- Maintains bone density (synergistic with estrogen)

30–70 ng/dL	1–5 pg/mL	0.5–2 mg/day	5–10 mg/wk
Target Total T	Target Free T	Cream dose	Injection

Application sites for testosterone cream: Clitoris, inner labia, inner arm, or upper thigh. Clitoral application maximizes local effect on sexual sensitivity.

SECTION 3: MENOPAUSE — STAGES AND MANAGEMENT

PERIMENOPAUSE vs. MENOPAUSE vs. POSTMENOPAUSE

Stage	Definition	Key Symptoms	HRT Approach
Perimenopause	Variable cycles, 2–10 yrs before menopause (avg 40–54)	Irregular periods, hot flashes, mood swings, sleep disruption	Low-dose transdermal estradiol + progesterone. Monitor every 3–6 mos.
Menopause	12 consecutive months without periods	Hot flashes, vaginal atrophy, sleep disturbance, mood changes	Standard HRT initiation — transdermal E2 + oral progesterone if needed.
Postmenopause	>12 months post-last period	Ongoing atrophy, bone loss, cardiovascular risk increase	Continued HRT if tolerated and no contraindications. Consider bone density monitoring.
Surgical Menopause	Oophorectomy at any age	Abrupt, severe — all symptoms at once	Immediate HRT initiation. Higher doses may be needed. Add testosterone if indicated.

SECTION 4: FEMALE SEXUAL HEALTH & PERFORMANCE

COMMON SEXUAL HEALTH CONDITIONS

HSDD (Hypoactive Sexual Desire Disorder)

Low or absent sexual desire causing distress. Affects 10–15% of all women and up to 40% of postmenopausal women. Most underdiagnosed sexual health condition in women.

GSM (Genitourinary Syndrome of Menopause)

Vaginal dryness, atrophy, thinning, burning, painful intercourse (dyspareunia), urinary urgency. Affects 50–60% of postmenopausal women. Entirely preventable and treatable.

Female Orgasmic Disorder

Difficulty achieving orgasm despite adequate arousal. Often due to inadequate stimulation, hormonal deficiency (testosterone), or clitoral hood adhesion.

Vaginismus / Dyspareunia

Painful intercourse — may be hormonal (GSM), musculoskeletal (pelvic floor dysfunction), or psychological. Pelvic floor PT + estrogen therapy highly effective.

SEXUAL HEALTH TREATMENTS & COMPOUNDS

Transdermal Testosterone	0.5–2 mg/day cream	First-line for low libido/HSDD. Strongest evidence of any agent.
PT-141 (Vyleesi)	1.75 mg SC 45 min prior	FDA approved for HSDD. Increases desire via brain melanocortin receptors.
Vaginal Estradiol	10 mcg suppository 2x/wk	GSM treatment. Minimal systemic absorption. Safe even in women with cancer history.

DHEA (Intrarosa)	6.5 mg vaginal daily	FDA approved for dyspareunia. Converts locally to E + T.
Ospemifene (Osphena)	60 mg oral daily	SERM — activates vaginal estrogen receptors without systemic estrogen.
Sildenafil (Addyi adjacent)	Off-label	Some benefit in postmenopausal FSAD — increases genital blood flow.
Oxytocin	Nasal spray	Bonding, arousal, orgasm facilitation — off-label but gaining evidence.
Bupropion	150–300 mg/day	Antidepressant with pro-sexual effects — counters SSRI-induced sexual dysfunction.

PRACTICAL SEXUAL WELLNESS — EVIDENCE-BASED TECHNIQUES

- Arousal takes longer as estrogen declines — allow 20–30 min of non-goal-oriented stimulation before penetration.
- Clitoral stimulation is primary for 70–80% of women — direct clitoral focus during partnered sex dramatically improves orgasm rates.
- Pelvic floor exercises (Kegels): 3 sets of 10–15 contractions daily — improve orgasm intensity and urinary control.
- Vibrator use (clitoral) is evidence-based for orgasmic disorder — endorsed by AASECT and ACOG.
- Vaginal moisturizers (Replens, hyaluronic acid) used 3x/week reduce dryness independent of HRT.
- Lubricants during intercourse: use water-based or silicone-based; avoid glycerin-containing products.
- Communication: explicit discussion of preferences, speeds, and stimulation types improves sexual satisfaction more than any compound.
- Mindfulness-based sex therapy: 4–8 sessions significantly improve desire, arousal, and satisfaction in women with HSDD.
- Exercise: 150+ min/week aerobic activity improves sexual function, body image, and hormone levels.
- Sleep: 7–9 hours — poor sleep reduces testosterone by up to 15% in women.

SECTION 5: RECOMMENDED BLOODWORK PANEL

BASELINE FEMALE HORMONE PANEL

Marker	Optimal Range	Frequency
Estradiol (E2)	Follicular: 30–100 pg/mL; Luteal: 100–300 pg/mL; Post-menopausal: 10–40 pg/mL (on HRT)	Every 6 months (on HRT)
Total Testosterone	30–70 ng/dL	Every 3 months
Free Testosterone	1–5 pg/mL	Every 6 months
SHBG	18–144 nmol/L (lower = more free T)	Every 6 months

Progesterone	Luteal: 5–20 ng/mL; Post-meno: <1	Every 3 months on HRT
DHEA-S	Age-appropriate (95–270 mcg/dL at 40+)	Annually
FSH	Post-meno: >30 mIU/mL confirms menopause	Once to confirm menopause
LH	Assists menopause confirmation	Once
Thyroid (TSH + Free T3/T4)	TSH 1–2 mIU/L optimal	Annually
CBC + CMP	Full metabolic baseline	Annually
Lipid Panel	LDL <100, HDL >60 optimal in women	Annually on HRT
Bone Density (DEXA)	T-score > -1.0	Every 2 years postmenopause

SECTION 6: FBF RECOMMENDED FEMALE PROTOCOLS

Perimenopause Starter Protocol

Estradiol patch 0.05 mg/day + Oral progesterone 100 mg at bedtime + Testosterone cream 1 mg/day to clitoral/labial area. Recheck in 8–12 weeks.

Menopause / Postmenopause Full Protocol

Estradiol patch 0.075–0.1 mg/day + Oral progesterone 200 mg at bedtime + Testosterone cream 1–2 mg/day + Vaginal estradiol 10 mcg 2x/week if GSM present.

HSDD / Low Libido Protocol

Testosterone cream 1–2 mg/day clitoris/inner thigh + PT-141 1.75 mg SC 45 min before activity (as needed) + Optimize estradiol if menopausal.

Surgical Menopause Protocol

Immediate high-dose estradiol (0.1 mg patch or 2 mg gel) + Progesterone 200 mg (if uterus present) + Testosterone 2 mg/day + Monitor in 6–8 weeks.

Active / Athletic Women

Testosterone cream 1–2 mg/day for performance + Estradiol for bone/recovery + DHEA 25 mg/day for energy + BPC-157 500 mcg if injury present.

Anti-Aging / Longevity Protocol

Estradiol + Progesterone HRT (as above) + GHK-Cu 1 mg SC 5x/week for skin/collagen + GH peptides (Sermorelin 300 mcg bedtime) + DHEA 25 mg/day.

IMPORTANT: All hormone therapies require a physician's evaluation, laboratory testing, and prescription in most countries. The protocols described here reflect current evidence-based practice but must be individualized by a qualified HRT specialist. This document is educational only and does not constitute medical advice. forGEDbyfreedom.com